

INDUSTRIAL AUTOMATION PRODUCT DEVELOPMENT: Build Versus Buy



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Industrial automation products cover a gamut of applications such as remote monitoring, programmable logic controllers (PLCs), human machine interface (HMI), robotics and so on. Let us explore and consider the pros and cons for embedded development platforms that can be used in building these products.

Cost and labour optimisation has created a huge demand for industrial automation products in a variety of industries. Apart from tangible reduction in manufacturing and labour costs, replacement of human labour with precise and rugged control systems also adds reliability to the manufacturing of goods and products. Industrial automation products are application- and industry-specific. Owing to the specificity of products' use, their sales volume can be as low as 20,000 to 30,000 per year.

Challenge for OEMs

Usually, embedded development involves high non-recurring engineering and

manufacturing costs, so low sales volume creates a non-optimal cost structure. Economies of scale cannot take effect due to low sales volume, so the original equipment manufacturers (OEMs) have to bear higher overhead costs.

Long product life. Although OEMs carry out multiple sourcing of components, long life of industrial automation products adds burden to the supply chain. Component obsolescence has to be handled throughout the product life of ten years or more.

Industrial automation includes products such as PLCs, HMI, robotic arms and more. These can be used in a variety of industries including marine, automotive, medical and food processing, among others. As the products are used for ten years or more, these should be supported for an extended period.

Embedded development: build versus buy

Chip based development or full custom development offers flexibility to OEMs to build products as per their budget and application requirements. However, owing to low sales volume, chip based development is not an ideal option for achieving cost optimisation. Full custom designs are preferable only if sales volumes exceed 70,000 to 80,000 per year.

Since industrial automation products are usually application- and industry-specific, sales volumes are not high enough to recover the investments in engineering and manufacturing. In order to generate positive economic returns, OEMs prefer to use the platform for longer duration to increase overall sales of the platform. However, this strategy has an issue.

Technological advancements happen at a faster pace and, thus, a platform

Fig. 1: Computer-on-module

